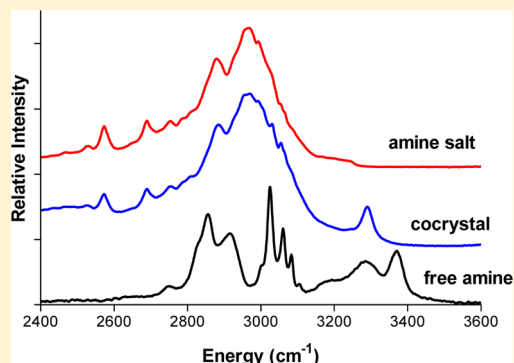


Cocrystal Systems of Pharmaceutical Interest: 2011

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ABSTRACT: The literature published during 2011 whose subject matter encompasses the cocrystallization of organic compounds having particular interest to pharmaceutical scientists has been summarized in an annual review. The papers cited in this review were drawn from the major physical, crystallographic, and pharmaceutical journals. After a brief introduction, the review is divided into sections that cover articles of general interest, the preparation of cocrystal systems and methodologies for their characterization, and more detailed discussion of cocrystal systems containing pharmaceutically relevant compounds. The review ends with a discussion of the draft Guidance for Industry document regarding the regulatory classification of pharmaceutical cocrystals that was issued at the end of 2011 by the Center for Drug Evaluation and Research (CDER) of the United States Food and Drug Administration.



1. INTRODUCTION

The literature published during 2011 continues to document how pharmaceutical scientists seek to use cocrystallization as a means to improve the oftentimes undesirable physical properties of drug substances undergoing development. The progress of this work has been documented in a series of review articles^{1–3} and in a series of reviews devoted to the literature of a particular time period.^{4–7} In the present review, the definition of a cocrystal proposed by Aakeröy will be used, namely, where cocrystal formation from supramolecular synthons is to be considered as forming from discrete neutral molecular species that are solids at ambient temperatures, and where the cocrystal is a structurally homogeneous crystalline material that contains the building blocks in definite stoichiometric amounts.⁸

A comprehensive overview of pharmaceutically interesting cocrystals has been published, which contained strong discussions of their physicochemical properties, design and isolation strategies, and characterization techniques.⁹ The article also contained summaries of pharmaceutically relevant cocrystals of carbamazepine, indomethacin, and ibuprofen as illustrative examples. Myerson and co-workers have published a review on the crystallization of pharmaceutically important compounds (including their cocrystals) that provides guidance as to how one might go about scaling up to industrial scale.¹⁰ Finally, as part of a more comprehensive review on the analysis of pharmaceutical polymorphs, the range of solid-state analytical techniques appropriate for the characterization of cocrystal systems has been reviewed.¹¹

As in previous reviews, primary attention will be paid to cocrystal systems for which there is a direct pharmaceutical interest, although papers having particular significance to the field will be discussed as well. The literature cited in the present review has been drawn from the major physical, crystallographic, and pharmaceutical journals, and consequently the coverage is not represented as being encyclopedic or comprehensive.

Apologies are presented in advance to any scientist in the field whose works have been inadvertently omitted.

2. ARTICLES OF GENERAL INTEREST

Cocrystal research is certain an exploration of crystal engineering, and Thomas has contributed an interesting summary of some of the early work that has brought the field to where it is.¹² After reading this article, one should then proceed to the article summarizing some recent developments in crystal engineering that have been made by scientists working in Asian countries that discusses the role of strong and weak interactions, the existence of entities and clusters in crystals, and the functionalities that can be achieved through the use of cocrystals.¹³

Since the phenomenon of hydrogen bonding strongly influences the crystal structure of a substance, the commentary provided by Desiraju on the recent IUPAC definition of the hydrogen bond is most useful.¹⁴ After citing the preamble to the IUPAC definition, “the hydrogen bond is an attractive interaction between a hydrogen atom from a molecule or a molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or different molecule, in which there is evidence of bond formation”, Desiraju proceeds to critically comment on the aspects of the new definition that have particular interest to those working in crystal engineering.

Desiraju has also written a detailed discussion of the nomenclature and definitions of hydrogen-bonding as a function of the strength of the bonds involved, pointing out that difficulties exist with the categorization of some of the weaker bonding types.¹⁵ Delving into the IUPAC definition in more depth, Desiraju points out that theory and experiment are given

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equal status, thus allowing empirical evidence of hydrogen bonding to enter into an analysis. He then goes on to list a number of criteria that would be useful as evidence, and provides some of the characteristics inherent to hydrogen bonds. Perhaps the most useful discussions in this paper are the footnotes to definition, criteria, and characteristics of hydrogen bonds, as here Desiraju critically evaluates various aspects of the new IUPAC definition.

The theoretical prediction of crystal structures of salts and cocrystals is of great interest, and Price and co-workers have demonstrated that identifying the position of protons involved in hydrogen-bonding is important to calculating the relative stabilities of structures, and have also concluded that the old pK_a difference rule is insufficient for confident assignment of an acidic proton position.¹⁶ The identification of supramolecular synthons is of great importance in crystal structure interpretation, and the transferability of multipole charge density parameters has been investigated to determine if they could be treated as modules across differing structures.¹⁷

Seaton has examined how one could use trends and differences in Hammett substituent constants as a means to predict the possibility of cocrystallization for two acids, reporting that the larger the difference in Hammett constants the more likely one is to obtain a cocrystal.¹⁸ This trend was ascribed to the increased degree of binding energy of the heteronuclear synthon that existed if the constants differed by an appreciable amount relative to the binding energies of the separate homonuclear synthons. In a systematic analysis of structures in the Cambridge Structural Database, it has been shown that molecular volume, shape, and flexibility are important properties that influence whether one may obtain cocrystals containing more than molecule per asymmetric unit.¹⁹

One of the driving forces causing pharmaceutical scientists to actively investigate cocrystal systems as new forms of drug substances is the promise of enhanced solubility of compounds whose solubility is less than that needed for an acceptable dosage form. It has been proposed that when a cocrystal of a drug substance does exhibit an enhanced solubility that persists for several hours that the phenomenon is similar to the metastable supersaturation state that can be achieved upon dissolution of amorphous substances.²⁰ Of course, the enhanced solid-state stability of cocrystallized products relative to amorphous forms is a clear advantage inherent to cocrystals. The dissolution of an acetaminophen/theophylline cocrystal has been compared to that of a simple physical mixture, and the faster dissolution rate of the cocrystal was confirmed.²¹ However, a solubility advantage could not be maintained for the theophylline component as precipitation of the less soluble monohydrate form was observed to take place.

Rodriguez-Hornedo and co-workers have investigated how micellar solubilization can be used as a tool in crystal engineering to optimize thermodynamic stability and eutectic points,²² and solubility, stability, and pH_{MAX} .²³ Since the solution composition at eutectic points is one of the factors defining the stability of the system, a model based on the ionization condition of the components was developed that would relate these properties to the presence of surfactants and solution pH. For example, it was found that the solubility and pH_{MAX} of carbamazepine cocrystals in micellar solutions of sodium dodecyl sulfate could be predicted by the models, and that the predictions were in agreement with experimental results.

The study of model cocrystal systems is of great value in establishing an information base for the understanding of more complicated systems. A number of cocrystals of benzamide with substituted benzoic acids have been structurally characterized, and a correlation between interaction energies and Hammett substitution constants was found.²⁴ The ability of several phenylalkylamines to form cocrystals with their respective chloride salts has been studied, and the infrared absorption of the products used to develop spectroscopic selection rules for proving (or disproving) the existence of a salt-cocrystal product.²⁵

The existence of stereoselectivity was observed in the salt-cocrystals of α -methyl-benzylamine, as the cocrystal could only be formed if the chloride salt and its free base were of opposite absolute configuration. The scope of polymorphic, solvatomorphic, and cocrystal products formed by orcinol (5-methyl-1,3-dihydroxybenzene) has been exhaustively studied after interaction of this compound with 15 different coforming agents.²⁶ A search for polymorphism in the cocrystals formed by pyrazinamide with six benzenecarboxylic acids has been conducted under a variety of interaction conditions (solvent-drop grinding, slurry, solution, and melt crystallization), but only a single crystal form was obtained for each product.²⁷

A different type of salt-cocrystal has been reported, namely, where the pharmaceutical agent is cocrystallized with a salt.²⁸ To demonstrate the principle, a series of ionic cocrystals were obtained that contained calcium chloride in conjunction with either barbituric acid, diacetamide, malonamide, nicotinamide, or piracetam. Depending on the compound under study, products could be obtained by direct crystallization from solution, as well as by slurry or solid-state processing methods. The products were all found to contain water of crystallization as a requisite part of the lattice structure.

While many cocrystal investigations have been concerned with the classical scope of synthon donors and acceptors, the use of halogen groups in supramolecular synthons is being investigated. The importance of electrostatic and geometric complementarity has been discussed for synthon combinations containing a combination of halogen bonds and hydrogen bonding.²⁹ This situation was brought to light owing to the fact that 2-point contacts are characteristic of hydrogen bonds, while 1-point interactions are associated with halogen...lone pair synthons. The ability of perfluorosuccinic acid to alter its molecular conformation relative to its hydrocarbon analogue has suggested that fluorination could be a general means to modify the shape of a coformer without changing its size.³⁰ These principles were illustrated through study of the structures of cocrystals containing caffeine and perfluorosuccinic or perfluoroadipic acids.

3. PREPARATION OF COCRYSTAL SYSTEMS, AND METHODOLOGIES FOR CHARACTERIZATION

It is certainly possible to produce cocrystals by evaporation from concentrated solutions, and this procedure works best if the coformers exhibit comparable degrees of solubility in the crystallizing solvent. In order to better predict the miscibility of a drug substance and a potential coformer, the use of Hansen solubility parameters has been investigated.³¹ Using indomethacin as a model compound the parameters of over 30 coformers were calculated, and then the difference in parameters between the drug and the coformers was calculated using established procedures. The predicted results were found

to be experimentally viable in nearly every instance, and, in addition, two new cocrystals were discovered after having been predicted.

A kinetically controlled crystallization process that entails rapid evaporation of the solvent from a solution containing the potential cofomers has been proposed as a rapid method for the screening of new cocrystals.³² Not only was use of the procedure able to yield a number of cocrystal products of several drug substances and potential cofomers, but the rapidity of formation should also facilitate the detection of metastable polymorphic forms of the products. The use of nonequilibrium conditions has also been used to obtain preferential enantiomeric enrichment during the cocrystallization of racemic phenylalanine and fumaric acid.³³

The cocrystallization of caffeine with glutaric acid from acetonitrile has been monitored using infrared absorption spectroscopy (attenuated total reflectance sampling) and particle vision measurement as means to effect feedback control over the process.³⁴ By controlling the crystallization parameters, it was shown that one could eliminate nucleation of an undesirable metastable crystal form and produce large particles with a minimum content of fines. The use of membrane-based crystallization technology has been investigated for the production of cocrystals of carbamazepine and saccharin.³⁵ In this approach, as long as the initial composition of the aqueous ethanol solvent system was optimized, the membrane technology enabled one to control the degree of supersaturation during the process and thus obtain the desired product.

There is little doubt that the use of solid-state grinding of the reactants in the presence of small quantities of solvent is a superior method to produce cocrystal products on the small scale,³⁶ although the scaling up of this methodology is not straightforward. Nevertheless, the use of a modified planetary mill with the capacity to process 48 samples in parallel has been investigated for the carbamazepine/saccharin, caffeine/oxalic acid, and caffeine/maleic acid cocrystal systems.³⁷ The use of cofomer milling prior to spontaneous cocrystal formation has been investigated for a number of known systems, where the reactants were initially milled to a particular particle size range and then allowed to form cocrystals in a solid-state convection mixing apparatus.³⁸ Reaction via eutectics or amorphous solids was shown not to be important to the process, and the fact that rates of cocrystal formation were most rapid for the smallest particle size fractions (i.e., 20–45 μm) was ascribed to increases in particle contact areas.

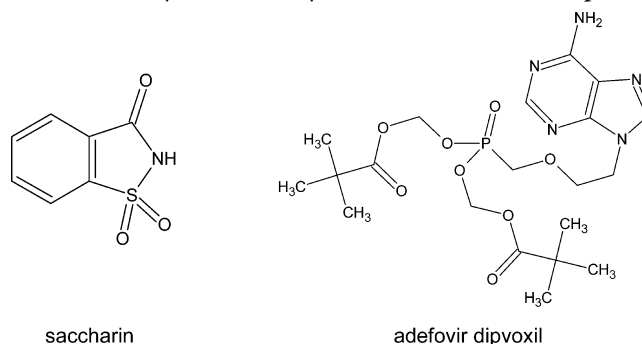
The rate of carbamazepine and nicotinamide cocrystal formation has been found to be accelerated by the enhanced water sorption of polyvinylpyrrolidone in the reaction mixture.³⁹ The mechanism for transformation of the drug/coformer/polymer ternary mixture was seen to proceed through moisture absorption by the polymer that was followed by dissolution of the components and formation of the cocrystal product. The efficient formation of the cocrystal product was explained by the increased mobility of water in the ternary mixture that led to a more effective dissolution and supersaturation of the cofomers. In addition, the polymer was found to alter the eutectic point associated with the carbamazepine/nicotinamide cocrystal, crystalline carbamazepine dihydrate, and solution phase system such that the thermodynamic stability of the cocrystal could be enhanced relative to the stability of the individual components.

Electrochemically induced reactions have been shown to afford a possible pathway for the preparation of cocrystal products, where the principle was established using a system consisting of cinnamic acid and 3-nitrobenzamide.⁴⁰ Cinnamate anions were neutralized by electrolytically generated hydrogen ions, whereupon the newly formed cinnamic acid was able to form a cocrystal product with the electrochemically inactive 3-nitrobenzamide. The methodology was proposed to the product removal of ionizable compounds at conditions for which conventional methods of crystallization were not practical.

4. COCRYSTAL SYSTEMS HAVING PHARMACEUTICAL INTEREST

The expanding literature of 2011 demonstrates the degree that cocrystal systems have taken the interest of pharmaceutical scientists in their continuing investigations for novel solid-state forms of active pharmaceutical ingredients. The following section of this review will concern discussions of published work conducted on cocrystal systems that are of pharmaceutical interest.

The 1:1 cocrystal formed by saccharin with adefovir dipvoxil:

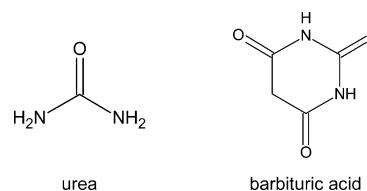


saccharin

adefovir dipvoxil

has been found to be more stable and exhibit superior dissolution relative to the drug substance alone.⁴¹ Diffraction analysis of the cocrystal revealed that it crystallized in a triclinic space group, and it was reported that the phosphoryl group and imide synthons were connected by N–H $\cdots$ O hydrogen bonds. While adefovir dipvoxil Form-I was found to completely degrade in 18 days when heated at 60 $^{\circ}\text{C}$, the superiority of the cocrystal was evident in that it remained chemically stable for 47 days when heated at 60 $^{\circ}\text{C}$.

The crystal structures of two polymorphic forms of the urea cocrystal with barbituric acid:

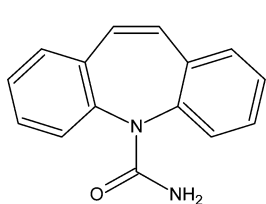


urea

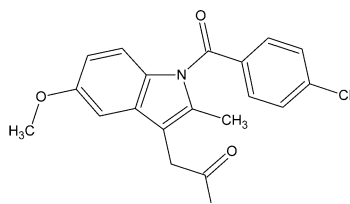
barbituric acid

have been obtained in order to confirm that barbituric acid adopts different mesomeric forms in the two polymorphs and to study the pattern of hydrogen-bonding in each.⁴² The two forms were both found to crystallize in monoclinic space groups ($P2_1/c$ for Form-I, and Cc for Form-II), with cocrystallization causing the barbituric acid to exhibit displaced charge density toward tautomeric forms of higher stability.

Even though carbamazepine is one of the most studied cocrystal formers, new reports continue to be published. In one work, a 1:1 cocrystal of carbamazepine with indomethacin



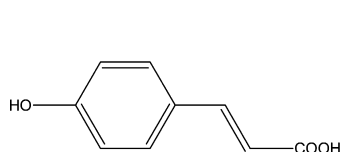
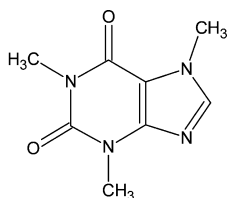
carbamazepine



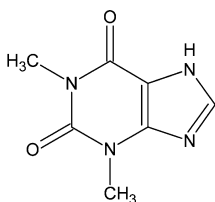
indomethacin

was produced by a milling process followed by exposure to 40 °C and 75% relative humidity for 21 days, and also by grinding in a mortar.⁴³ The product was characterized by X-ray powder diffraction, and the resulting pattern was indexed to a monoclinic unit cell. In another study, a metastable, monotropic, polymorph of the carbamazepine/nicotinamide cocrystal was produced by isothermal crystallization from the glassy state, and critically studied by means of rapid-heating differential scanning calorimetry.⁴⁴

The structures of a number of cocrystals of the nutraceutical compound *p*-coumaric acid with caffeine and theophylline:

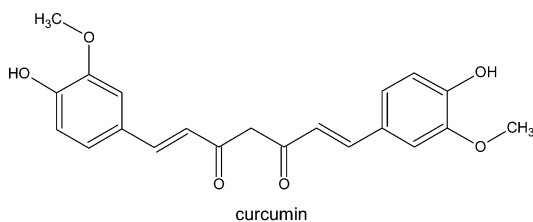
*p*-coumaric acid

caffeine

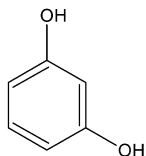


theophylline

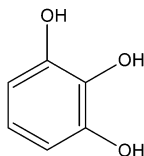
have been obtained, namely, the 1:1 and 1:2 stoichiometric cocrystals with caffeine and two polymorphs of the 1:1 cocrystal with theophylline.⁴⁵ While both theophylline cocrystals exhibited imidazole-carboxylic acid synthons, one polymorph also contained a carbonyl-hydroxyl synthon, and the other contained an imidazole-hydroxyl synthon. In another study, caffeine was found to form a 1:1 cocrystal with (+)-catechin, a 1:1 cocrystal with (−)-catechin-3-*O*-gallate, and a 1:1:2 (+)-catechin/(−)-epicatechin/caffeine cocrystal.⁴⁶



curcumin



resorcinol

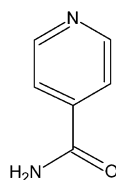


pyrogallol

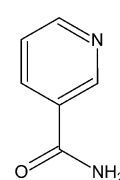
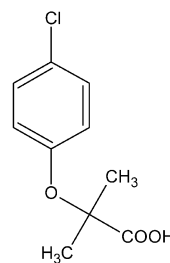
The poor aqueous solubility and dissolution of curcumin (the principle curcuminoid of the Indian spice tumeric) has been improved by cocrystallization with resorcinol and pyrogallol.⁴⁷

The apparent solubility of the curcumin/resorcinol cocrystal was estimated as being 4.7 times higher than the solubility of curcumin Form-I, and the apparent solubility of the curcumin/pyrogallol cocrystal was estimated as being 11.8 times higher. These solubility enhancements were found to translate into greatly improved dissolution rates for the cocrystals relative to curcumin itself.

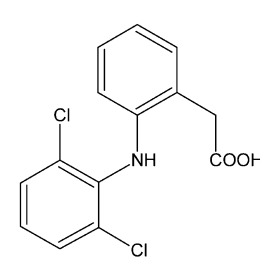
During a study of the isonicotinamide cocrystallization with vitamin B3 (nicotinamide), clofibric acid, and diclofenac:



isonicotinamide

vitamin B3
(nicotinamide)

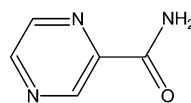
clofibric acid



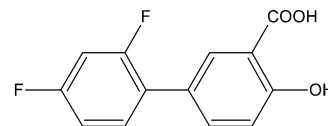
diclofenac

it was found that not only could 1:1 cocrystals be formed by isonicotinamide with clofibric acid and diclofenac, but that isonicotinamide would form a cocrystal with its positional isomer, vitamin B3.⁴⁸ In this work, the cocrystal forming ability of nicotinamide and isonicotinamide was investigated through the density functional theory calculations.

The 1:1 cocrystal formed by pyrazinamide and diflunisal:



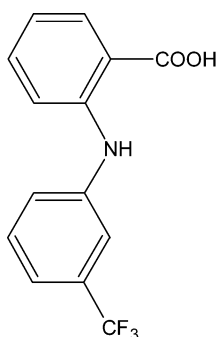
pyrazinamide



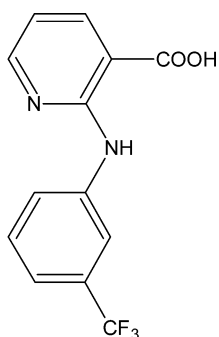
diflunisal

was only able to be formed by grinding equimolar amounts of the reactants followed by thermal treatment at 80 °C.⁴⁹ The cocrystal was also obtained by means of ethanol-assisted ball mill grinding and by room temperature annealing of the mixture obtained by neat ball mill grinding. The dual-drug product was described as being of value in that side effects of pyrazinamide could be mitigated and that the aqueous solubility of diflunisal could be improved.

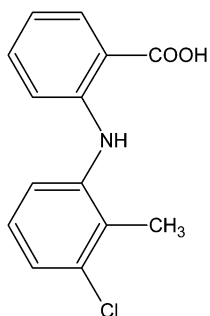
The structures of the cocrystals formed by nicotinamide with several fenamic acids:



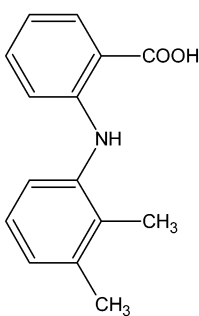
flufenamic acid



niflumic acid



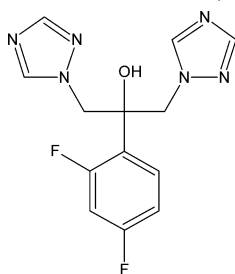
tolfenamic acid



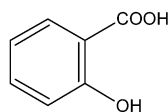
mefenamic acid

have been reported, with two being obtained in the monoclinic $P2_1/c$ space group and two in the triclinic $P\bar{1}$ space group.⁵⁰ Despite the fact that the four cocrystals each formed using the intramolecular $N-H\cdots O=C$ heterosynthon, differences in hydrogen-bonding patterns led to the existence of differences in stability among the products.

The structure of a 1:1 cocrystal of fluconazole with salicylic acid:



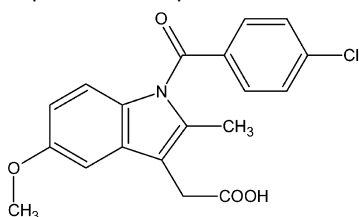
fluconazole



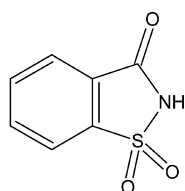
salicylic acid

has been reported, with this product crystallizing in the triclinic $P\bar{1}$ space group.⁵¹ In this structure, the fluconazole and salicylic acid molecules are each joined by hydrogen bonds into homomeric centrosymmetric dimers, whereupon these dimers are further linked by an additional $O-H\cdots N$ hydrogen bond (between one of the salicylate carboxylic acid OH groups and a nitrogen atom on a fluconazole triazole).

The solubility behavior and solution-phase chemistry of the cocrystal formed by saccharin with indomethacin:



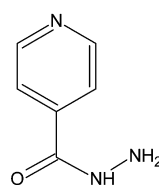
indomethacin



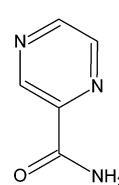
saccharin

has been studied in methanol, ethanol, and ethyl acetate, with the generation of phase solubility diagrams.⁵² It was found that the solubility of the cocrystal decreased with increasing concentration of saccharin, which could be explained in terms of the solubility product and solution-phase complexation.

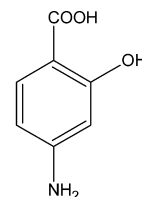
Structures of the 1:1 cocrystals formed by 4-aminosalicylic acid with isoniazid and pyrazinamide:



isoniazid



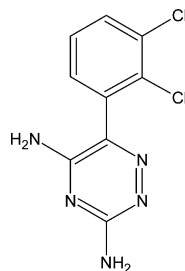
pyrazinamide



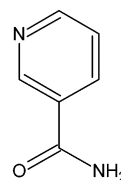
4-aminosalicylic acid

have been reported, with hydrogen bonding involving $COOH\cdots N$ pyridine synthons.⁵³ Interestingly, in one of the cocrystals, only partial proton transfer existed in one of the hydrogen bonds, and the extent of proton transfer was found to depend on temperature. In another study, the carbohydrazide functional group of isoniazid was reacted with a series of ketones, and the effect of this modification on the cocrystal formation with 3-hydroxybenzoic acid was evaluated.⁵⁴

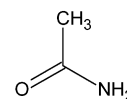
Cocrystal products were obtained through the interaction of nicotinamide and acetamide and with lamotrigine:



lamotrigine



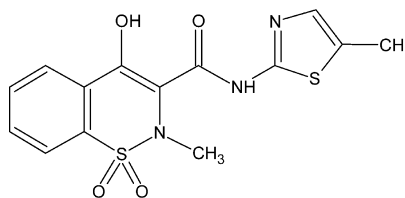
nicotinamide



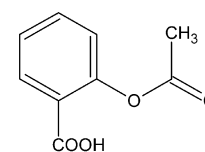
acetamide

while salts were obtained when the drug substance was reacted with 4-hydroxybenzoic acid, acetic acid, and saccharin.⁵⁵ The enthalpy of formation associated with the salt forms was found to be larger than the enthalpies obtained for the cocrystals, although this difference in stability did not directly translate into a solubility trend. In fact, dissolution of the two cocrystal products resulted in formation of a lamotrigine hydrate.

Solution-phase crystallization, tetrahydrofuran slurring, or solvent-assisted grinding has been used to obtain a cocrystal of meloxicam and aspirin:⁵⁶



meloxicam

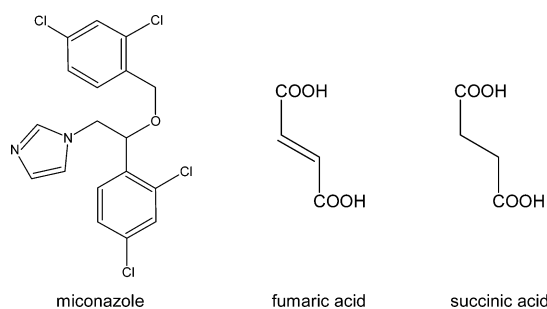


aspirin

Aspirin was chosen as the coformer owing to its desired physicochemical and pharmacokinetic properties, and the cocrystal was found to exhibit superior kinetic solubility and the potential to decrease the time for the meloxicam to reach the human therapeutic concentration.

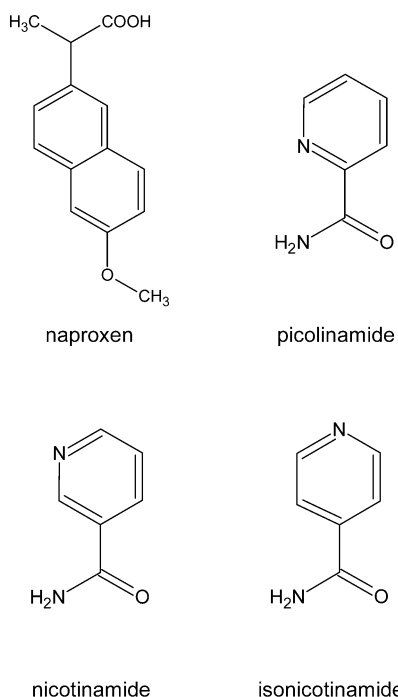
The ability of miconazole to form salts and cocrystals has been studied, and while a salt was obtained upon interaction with

maleic acid, cocrystal products were obtained with half-neutralized fumaric and succinic acids:⁵⁷



It was found that although formation of all products improved the dissolution rate of the drug substance, the drug substance in the maleate salt and in the hemifumarate cocrystal was not stable. Since the hemisuccinate cocrystal exhibited superior dissolution and stability, it was considered to be appropriate for further development.

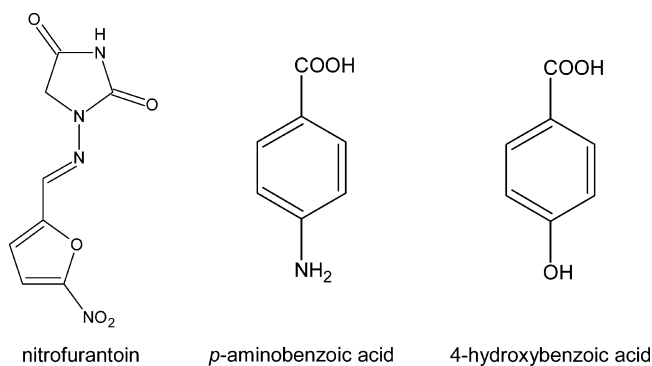
Using a hot-stage contact method for screening, cocrystals were obtained by the interaction of naproxen with three amide compounds:



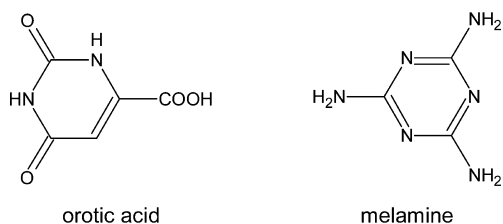
although no cocrystal product could be obtained with pyrazinamide.⁵⁸ The existence of a supramolecular synthon based on the O—Hcarboxylic acid...Naromatic hydrogen bond was found in the structures of all cocrystal products, and evidence for its presence was also detected in the respective infrared absorption spectra.

Nitrofurantoin is known to transform into a hydrated crystal form in aqueous media, but it has been reported that its cocrystals with *p*-aminobenzoic acid⁵⁹ and with 4-hydroxybenzoic acid⁶⁰ exhibit a superior range of physicochemical properties.

The superiority of these products was amply demonstrated, as when exposed to water, the *p*-aminobenzoic acid cocrystal exhibited minimal phase transformation to the hydrate and its dissolution rate was comparable to that of the drug substance itself. The 4-hydroxybenzoic acid was found to exhibit complete physical stability when exposed to accelerated test conditions, and was also found to be photostable.

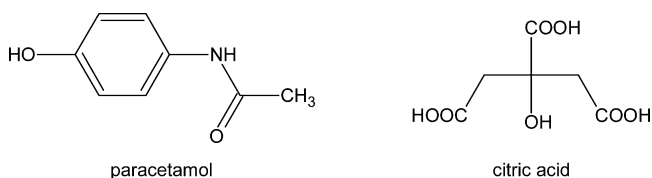


The structure of a hydrated cocrystal of melamine and orotic acid has been reported



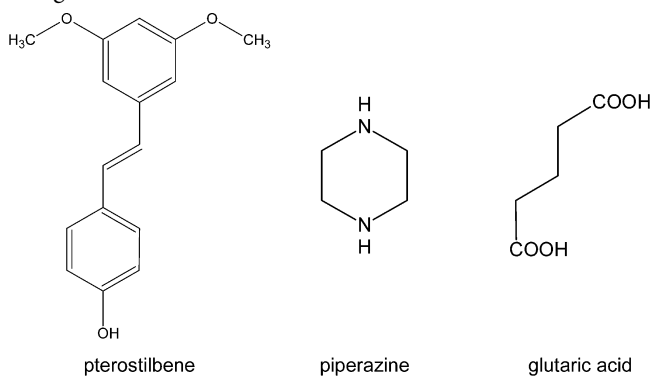
where it was learned through variable-temperature studies that fluctuation in the hydrogen atoms of the crystalline water played a key role in interesting dielectric phenomena.⁶¹ Large changes in the dielectric constant of the cocrystal were observed upon heating, which were related to dehydration and its effect on the hydrogen-bonding between molecular layers in the solid.

A 1:2 cocrystal of citric acid and paracetamol was obtained by a slow evaporation method,



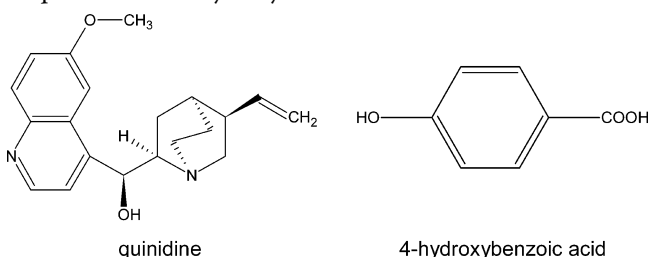
where the phenolic—OH of one paracetamol molecule acts as a donor in hydrogen-bonding to a carbonyl group on a citric acid molecule, while the phenolic—OH of the other paracetamol molecule acts as a hydrogen-bond acceptor from the quaternary C—OH of a citric acid molecule.⁶² The Raman spectra of the reactants and their resulting product were completely assigned, and trends in the spectra were used to confirm the existence of a cocrystal species.

The physical properties of pterostilbene have been greatly improved by the formation of cocrystal products with piperazine and glutaric acid:⁶³



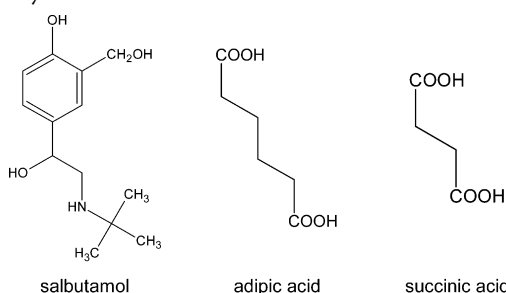
The aqueous solubility of the piperazine cocrystal was found to be approximately six times higher than the solubility of the drug substance itself, while the glutaric acid cocrystal was seen to rapidly disproportionate in water. Procedures were developed that enabled the cocrystal products to be obtained on the multigram scale.

A variety of investigational techniques have been used to evaluate the predictability of cocrystal formation in the instance of quinidine and 4-hydroxybenzoic acid:⁶⁴



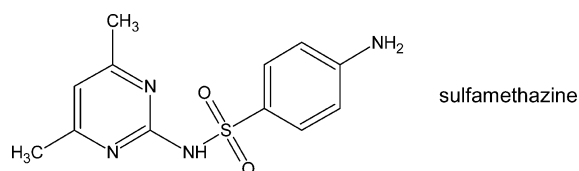
The product was crystallized in a monoclinic space group, and the structure was stabilized by a set of charge-assisted heterosynthons. The solid-state NMR spectrum of the cocrystal was assigned, with support being obtained by means of density functional theory calculations.

Structures of the nonsolvated cocrystals of salbutamol with adipic acid and succinic acid have been reported, as well as the tetra-methanolate solvatomorph of the salbutamol hemisuccinate cocrystal:⁶⁵



The intrinsic dissolution of the adipic acid cocrystal was found to be approximately four times lower than that of salbutamol itself, suggesting that the cocrystal could be used as an alternative to the more rapidly dissolving salbutamol sulfate currently used in dosage forms.

The crystal structures of a series of cocrystals were formed between sulfamethazine:



and 4-hydroxybenzoic acid, 2,4-dihydroxybenzoic acid, 3,4-dichlorobenzoic acid, sorbic acid, fumaric acid, 1-hydroxy-2-naphthoic acid, benzamide, picolinamide, 4-hydroxybenzamide, and 3-hydroxy-2-naphthoic acid have been reported, and the patterns of hydrogen bonding in each discussed in detail.⁶⁶ The structure of a 2:1 cocrystal of sulfamethazine and theophylline has also been reported, where each sulfamethazine molecule exists as a different tautomer in the crystal.⁶⁷

5. PHARMACEUTICAL COCRYSTALS: THE UNITED STATES FOOD AND DRUG ADMINISTRATION WEIGHS IN

In the last annual review,⁷ it was observed that although the potential benefits of using cocrystal products as active

pharmaceutical ingredients were recognized, the regulatory status regarding the use of cocrystals in pharmaceutical products was unresolved. The key question for development scientists was whether a cocrystal would be defined as a physical mixture (enabling its classification within current compendial guidelines) or as a new chemical entity requiring full safety and toxicology testing.

The Center for Drug Evaluation and Research (CDER) of the United States Food and Drug Administration has addressed this issue, and issued a draft Guidance for Industry document regarding the regulatory classification of pharmaceutical cocrystals at the end of 2011.⁶⁸ In this document, FDA has chosen to define cocrystals as “solids that are crystalline materials composed of two or more molecules in the same crystal lattice”. To differentiate salts from cocrystals, FDA defined the interaction among cocrystal coformers as being “in a neutral state” that “interact via nonionic interactions.” FDA went on to classify cocrystals within its current regulatory framework as “dissociable API-excipient molecular complexes (with the neutral guest compound being the excipient).” Because FDA has defined the molecular association of the drug substance and its excipient within a crystal lattice, FDA has taken the position that a cocrystal may be treated as a drug product intermediate.

According to the Guidance, in order for a cocrystal of a drug substance to be classified as an “API-excipient” molecular complex, a New Drug Application (or an Abbreviated New Drug Application) must contain the results of two studies. The first of these was stated as, “Determine whether, in the crystalline solid, the component API with the excipient compounds in the cocrystal exist in their neutral states and interact via nonionic interactions, as opposed to an ionic interaction, which would classify this crystalline solid as a salt form.” The consequence of this requirement is that in effect, applicants must provide evidence that no ionic interaction or proton transfer is part of the supramolecular synthon in the cocrystal. The second condition expressed in the Guidance is that the applicants must show that the drug substance dissociates from the coformer prior to the moment when the drug substance carries out its pharmacological function.

As one might imagine, publication of the draft Guidance led to a considerable amount of discussion during 2012. While it is beyond the scope of a 2011 annual literature review to encapsulate the discussion, it is to be noted that a significant discussion was held by research leaders during the Indo-US Bilateral Meeting on the Evolving Role of Solid State Chemistry in Pharmaceutical Science (Manesar, India), where an entire session was devoted to a panel discussion of the draft Guidance.⁶⁹ In addition, many comments on the draft Guidance have been submitted to FDA and published on their Web site,⁷⁰ including those provided by Abbott, AstraZeneca, Boeringer Ingelheim, Bristol-Myers Squibb, GlaxoSmithKline, Hoffman-LaRoche, Eli Lilly, Merck, Novartis, and Pfizer. Naturally the comments span a variety of viewpoints, with some linking definitions of cocrystals with solvatomorphs, and others linking definitions of cocrystals with salts.

A major problem with the draft guidance begins with the definition provided for cocrystals, “Solids that are crystalline compounds of two or more molecules in the same crystal lattice.” This highly general definition spurred a variety of viewpoints in the published responses, with some linking definitions of cocrystals with solvatomorphs, and others linking definitions of cocrystals with salts. As stated above, most workers in the field would agree with the superior definition of Aakeröy that

cocrystals are formed by the cocrystallization of neutral molecules that are solids at ambient temperatures.⁸

Nevertheless, the draft Guidance seeks to establish a black/white distinction that the agency would use to differentiate between salts and cocrystals. However, it is widely recognized that a “salt” and a “cocrystal” actually represent extremes in the degree of proton transfer, where whether a product is classified as a salt or a cocrystal depends on how effectively a proton can be moved from an acid to a base. While the FDA attempted to base its differentiation solely on differences in ionization constants, solid-state scientists recognize that patterns of hydrogen-bonding in a crystal will also play an important role during cocrystallization. Depending on the details of the crystal structure, the predicted outcome of two coformers (especially when ΔpK_a is between 2 and 3) could be a salt, a cocrystal, or some species exhibiting an intermediate degree of proton transfer.

The draft Guidance does demonstrate, however, that FDA is very aware that cocrystals will appear as active pharmaceutical ingredients in many regulatory filings, and that the agency is actively trying to determine how to handle the classification issues. FDA has faced similar issues before, having issued Guidance documents for polymorphs (and solvatomorphs) of drug substances, and for salt forms of active pharmaceutical ingredients. In their comments on the draft Guidance, Triclinic Labs succinctly summarized three possibilities open to FDA: (1) retract the draft Guidance and let cocrystals be regulated as salts, (2) modify the draft Guidance to classify cocrystals as product intermediates that do not require regulation, or (3) create a new Guidance document that is internationally harmonized with other regulatory agencies and scientific thought, and which will provide the necessary clarifications related to cocrystals.

AUTHOR INFORMATION

Notes

The authors declare no competing financial interest.

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